Program: **Diploma in Fire Technology and Safety**	
Course Code: **4469**	Course Title: **Strength of Materials Lab**
Semester: **4**	Credits: **1.5**
Course Category: **Program Core**	
Periods per week: **3 (L:0, T:0, P:3)**	Periods per semester: **45**

Course Objectives:

- To apply material testing principles to fire safety applications.
- To understand the mechanical properties of materials used in fire protection.
- To analyze the behavior of materials under different loading conditions.

Course Outcomes:

On completion of the course, the student will be able to:

COn	**Description**	**Duration (Hours)**	**Cognitive Level**
CO1	Conduct tension and impact tests on different materials.	9	Applying
CO2	Analyze the deflection behavior of beams and torsional properties of materials.	10	Applying
CO3	Evaluate the hardness and fatigue properties of fire-resistant materials.	10	Applying
CO4	Perform compressive strength tests on concrete and flexural strength tests on beams.	12	Applying
	Lab Exam	4	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3	2					
CO3	3	3					
CO4	3	3					

3-Stronglymapped,2-Moderatelymapped,1-Weaklymapped

Course Outline:

Module outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Conduct tension and impact tests on different materials.**		
M1.01	Conduct tension test on metal specimens.	4	Applying
M1.02	Perform Izod and Charpy impact tests	5	Applying
CO2	**Analyze the deflection behavior of beams and torsional properties of materials.**		
M2.01	Deflection test on cantilever and simply supported beams	6	Applying
M2.02	Conduct torsion test on metallic rods	4	Applying
	Lab Exam I	2	
CO3	**Evaluate the hardness and fatigue properties of fire-resistant materials.**		
M3.01	Perform Brinell and Rockwell hardness tests	5	Applying
M3.02	Evaluate fatigue behavior using fatigue testing machine.	5	Applying
CO4	**Perform compressive strength tests on concrete and flexural strength tests on beams.**		
M4.01	Compressive test on concrete cubes.	6	Analyzing
M4.02	Flexural strength test on beams.	6	Analyzing
	Lab Exam–II	2	